Haotian Zhang

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Education

University of Southern California Starting Spring 2027
Incoming undergraduate in Applied and Computational Mathematics.

American University of Paris August - December 2026
Visiting Student · USC Fall Semester in Paris

Research Interests

Video Reasoning; World Models; Foundation Models; Applied Mathematics.

My interests mainly lie in AI and mathematics. I currently work on visual reasoning and world models. Specifically, my recent research focuses on physical reasoning in world models and its generalization across visual tasks. My work involves cognitively grounded task design, model training, and evaluation.

Honors and Awards

Hugging Face #1 Paper of the Day September 2026
For *Training Object Permanence in World Models*.

United States of America Mathematical Olympiad (USAMO) 2024
Qualifier; American Invitational Mathematics Examination (AIME): 13/15.

American Mathematics Competition (AMC 12) 2024
Global Top 1%.

Berkeley Math Tournament 2023
National High Distinction (individual); national Top 2 overall and Top 3 in the Guts round.

S.-T. Yau High School Science Award - Mathematics 2024
Second Prize (Top 8), Southern Region Finals.

Publications

* Equal contribution. Ellipses shorten long author lists without changing the original order.

Preprint

Training Object Permanence in World Models

Haotian Zhang*, Fengyuan Yu*, Dezhi Luo*, Haoran Sun, Zehong Zhao, Qingying Gao, ..., Yilun Du, Hokin Deng.

arXiv:2609.28654, 2026. Co-first author.

Paper | Project | Code

Research Experience

Hippocampal June 2026 - Present
Research Intern

Training Object Permanence in World Models (WROP) June - September 2026
Co-first author, core team member

- Designed 150 Blender-based 3D task generators on object permanence and solidity, and scaled them to a 1.5-million-sample training corpus.
- Contributed to training a 16B-parameter video model and human evaluation.
- Co-first authored the paper, which ranked #1 on Hugging Face's Papers of the Day.

Research Experience (continued)

University of California, Santa Cruz June - August 2025
Research Intern, Science Internship Program (SIP)

- Developed a generative-AI tool to scaffold research-paper reading through simulated reader-AI dialogue and iterative refinement of system prompts.
- Designed and evaluated prompts to help AI systems explain complex academic texts.
- Evaluated NotebookLM's Audio Overview for vocal tone, pacing, and content accuracy against professional podcast standards; explored prompt strategies for more natural narration.

Cambridge Centre for International Research (CCIR) June - September 2025
Research Student

- Completed coursework in Applied Linguistics & Big Data.
- Authored a research paper on the social functions of multilingualism along the ninth- and tenth-century Silk Road, combining linguistic theory with data-informed analysis.